Example: Kalman Filter
Reference: Thrun, Burgard, Fox, "Probabilistic Robotics," MIT Press 2006

The representation of a Gaussian by its mean and covariance is called the *moments representation*. This is because the mean and covariance are the first and second moments of a probability distribution; all other moments are zero for normal distributions. In this chapter, we will also discuss an alternative representation, called *canonical representation*, or sometimes *natural representation*. Both representations, the moments and the canonical representations, are functionally equivalent in that a bijective mapping exists that transforms one into the other (and back). However, they lead to filter algorithms with orthogonal computational characteristics.

This chapter introduces the two basic Gaussian filter algorithms.

- Section 3.2 describes the Kalman filter, which implements the Bayes filter using the moments representation for a restricted class of problems with linear dynamics and measurement functions.
- The Kalman filter is extended to nonlinear problems in Section 3.3, which describes the extended Kalman filter.
- Section 3.4 describes the information filter, which is the dual of the Kalman filter using the canonical representation of Gaussians.

3.2 THE KALMAN FILTER

3.2.1 Linear Gaussian Systems

Probably the best studied technique for implementing Bayes filters is the *Kalman filter (KF)*. The Kalman filter was invented in the 1950s by Rudolph Emil Kalman, as a technique for filtering and prediction in linear systems. The Kalman filter implements belief computation for continuous states. It is not applicable to discrete or hybrid state spaces.

The Kalman filter represents beliefs by the moments representation: At time t, the belief is represented by the the mean μ_t and the covariance Σ_t. Posteriors are Gaussian if the following three properties hold, in addition to the Markov assumptions of the Bayes filter.

1. The next state probability $p(x_t \mid u_t, x_{t-1})$ must be a *linear* function in its arguments with added Gaussian noise. This is expressed by the following equation:

$$x_t \quad = \quad A_t x_{t-1} + B_t u_t + \varepsilon_t \; . \tag{3.2}$$

Here x_t and x_{t-1} are state vectors, and u_t is the control vector at time t. In our notation, both of these vectors are vertical vectors, that is, they are of the form

$$x_t = \begin{pmatrix} x_{1,t} \\ x_{2,t} \\ \vdots \\ x_{n,t} \end{pmatrix} \quad \text{and} \quad u_t = \begin{pmatrix} u_{1,t} \\ u_{2,t} \\ \vdots \\ u_{m,t} \end{pmatrix} . \tag{3.3}$$

A_t and B_t are matrices. A_t is a square matrix of size $n \times n$, where n is the dimension of the state vector x_t. B_t is of size $n \times m$, with m being the dimension of the control vector u_t. By multiplying the state and control vector with the matrices A_t and B_t, respectively, the state transition function becomes *linear* in its arguments. Thus, Kalman filters assume linear system dynamics.

The random variable ε_t in (3.2) is a Gaussian random vector that models the randomness in the state transition. It is of the same dimension as the state vector. Its mean is zero and its covariance will be denoted R_t. A state transition probability of the form (3.2) is called a *linear Gaussian*, to reflect the fact that it is linear in its arguments with additive Gaussian noise.

Equation (3.2) defines the state transition probability $p(x_t \mid u_t, x_{t-1})$. This probability is obtained by plugging Equation (3.2) into the definition of the multivariate normal distribution (3.1). The mean of the posterior state is given by $A_t x_{t-1} + B_t u_t$ and the covariance by R_t:

$$\begin{aligned} & p(x_t \mid u_t, x_{t-1}) \\ &= \det(2\pi R_t)^{-\frac{1}{2}} \exp\left\{-\tfrac{1}{2}(x_t - A_t x_{t-1} - B_t u_t)^T R_t^{-1}(x_t - A_t x_{t-1} - B_t u_t)\right\} \end{aligned} \tag{3.4}$$

2. The measurement probability $p(z_t \mid x_t)$ must also be *linear* in its arguments, with added Gaussian noise:

$$z_t = C_t x_t + \delta_t . \tag{3.5}$$

Here C_t is a matrix of size $k \times n$, where k is the dimension of the measurement vector z_t. The vector δ_t describes the measurement noise. The distribution of δ_t is a multivariate Gaussian with zero mean and covariance Q_t. The measurement probability is thus given by the following multivariate normal distribution:

$$p(z_t \mid x_t) = \det(2\pi Q_t)^{-\frac{1}{2}} \exp\left\{-\tfrac{1}{2}(z_t - C_t\, x_t)^T\, Q_t^{-1}\, (z_t - C_t\, x_t)\right\} \tag{3.6}$$

1: **Algorithm Kalman_filter($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t$):**
2: $\bar{\mu}_t = A_t\ \mu_{t-1} + B_t\ u_t$
3: $\bar{\Sigma}_t = A_t\ \Sigma_{t-1}\ A_t^T + R_t$
4: $K_t = \bar{\Sigma}_t\ C_t^T (C_t\ \bar{\Sigma}_t\ C_t^T + Q_t)^{-1}$
5: $\mu_t = \bar{\mu}_t + K_t(z_t - C_t\ \bar{\mu}_t)$
6: $\Sigma_t = (I - K_t\ C_t)\ \bar{\Sigma}_t$
7: return μ_t, Σ_t

Table 3.1 The Kalman filter algorithm for linear Gaussian state transitions and measurements.

3. Finally, the initial belief $bel(x_0)$ must be normal distributed. We will denote the mean of this belief by μ_0 and the covariance by Σ_0:

$$bel(x_0) \quad = \quad p(x_0) \quad = \quad \det(2\pi\Sigma_0)^{-\frac{1}{2}}\ \exp\left\{-\tfrac{1}{2}\ (x_0 - \mu_0)^T \Sigma_0^{-1} (x_0 - \mu_0)\right\}$$

These three assumptions are sufficient to ensure that the posterior $bel(x_t)$ is always a Gaussian, for any point in time t. The proof of this non-trivial result can be found below, in the mathematical derivation of the Kalman filter (Section 3.2.4).

3.2.2 The Kalman Filter Algorithm

The Kalman filter algorithm is depicted in Table 3.1. Kalman filters represent the belief $bel(x_t)$ at time t by the mean μ_t and the covariance Σ_t. The input of the Kalman filter is the belief at time $t-1$, represented by μ_{t-1} and Σ_{t-1}. To update these parameters, Kalman filters require the control u_t and the measurement z_t. The output is the belief at time t, represented by μ_t and Σ_t.

In Lines 2 and 3, the predicted belief $\bar{\mu}$ and $\bar{\Sigma}$ is calculated representing the belief $\overline{bel}(x_t)$ one time step later, but before incorporating the measurement z_t. This belief is obtained by incorporating the control u_t. The mean is updated using the deterministic version of the state transition function (3.2), with the mean μ_{t-1} substituted for the state x_{t-1}. The update of the covariance considers the fact that states depend on previous states through the linear matrix A_t. This matrix is multiplied twice into the covariance, since the covariance is a quadratic matrix.